Carbohydrate Polymers 325 (2024) 121568

Contents lists available at ScienceDirect

Carbohydrate Polymers

journal homepage: www.elsevier.com/locate/carbpol

Hyaluronan-arginine enhanced and dynamic interaction emerges from distinctive molecular signature due to electrostatics and side-chain specificity

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ARTICLE INFO

Keywords:
Hyaluronan
Arginine
Molecular dynamics
NMR
Sugar–peptide interactions

ABSTRACT

Hyaluronan is a natural carbohydrate polymer with a negative charge that fosters gel-like conditions crucial for its cellular functions and industrial applications. As a recognized ligand for proteins, understanding their mutual interactions provides solid ground to tune hyaluronan's gel properties using biocompatible peptides. This work employs NMR and molecular dynamics simulations to identify molecular motifs relevant to hyaluronan–peptide interactions using arginine, lysine, and glycine oligopeptides. Arginine-rich peptides exhibit the strongest binding to hyaluronan according to chemical shift perturbation measurements, followed distantly by the similarly charged lysine. This difference highlights the significance of electrostatics and the peculiarities of the guanidinium side chain in arginine, capable of non-polar interactions that further stabilize the binding. Additional nuclear Overhauser effect measurements do not show stable interaction partners, precluding strong and well-defined complexes. Finally, molecular simulations support our findings and show an extended but significant interaction region, especially for arginine, responsible for the observed enhanced binding, which can also promote cross-linking of hyaluronan polymers. Our findings pave the way for optimizing biocompatible peptides to alter hyaluronan gels' properties efficiently and also explain why hyaluronan–protein interaction typically involves positively charged arginine-rich regions also capable of forming hydrogen bonds and non-polar interactions.

1. Introduction

Hyaluronan, or hyaluronic acid (HA), is a naturally occurring, negatively charged polyelectrolyte. It is composed of a repetitive disaccharide unit consisting of D-glucuronic acid (GlcA) and *N*-acetyl-D-glucosamine (GlcNAc) ($[-\beta(1,4)$-GlcA-$\beta(1,3)$-GlcNAc-$]_n$) (Toole, 2004). Hyaluronan is synthesized by specialized vertebrate cells in the plasma membrane and then secreted into the extracellular environment (DeAngelis, 1999). It can attain molecular weights of up to thousands of kDa or be enzymatically digested within the extracellular space to produce fragments of various sizes in a controlled manner (McAtee et al., 2014). Hyaluronan and other molecular components collectively form a carbohydrate-rich layer on the outer surface of cell membranes known as the glycocalyx (Tarbell & Cancel, 2016). Current studies indicate that secretion, concentration, and weight of HA polysaccharide chains in the glycocalyx correlate with wound healing (Aya & Stern, 2014), arthritis
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development (Zhang & Christopher, 2015), and even cancer resistance (Hou et al., 2022; Tian et al., 2013). In fact, it is well-established that HA actively participates in cell–cell communication, protein recognition, and disease development (Berdiaki et al., 2023).

The extracellular matrix is ubiquitous in connective tissues, where it plays a pivotal role in defining tissue elasticity and permeability (Cowman et al., 2015; Giubertoni, Burla, et al., 2019). Notably, it was a long-standing belief that HA had mostly a passive protective function providing, structural support and lubrication in these tissues (Fraser 润滑 et al., 1997). Such functions are often attributed to its uncertain gel-like structure where only its soft and highly dynamic nature is really measurable (van Oosten & Janmey, 2013). Although the precise structural nature of the extracellular matrix remains debated, it is certain that some of its constituents, including HA, can readily form hydrogels. These hydrogels have a wide range of medical and biotechnological applications, and HA hydrogels, in particular, are known for their

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https://doi.org/10.1016/j.carbpol.2023.121568
Received 6 October 2023; Received in revised form 1 November 2023; Accepted 5 November 2023
Available online 14 November 2023

糖萼是糖蛋白和糖脂覆盖物，其围绕在一些细菌，上皮细胞和其他细胞的细胞膜上。 大多数动物上皮细胞在其质膜的外表面上具有类似绒毛的涂层。 该涂层由几种膜糖脂和糖蛋白的碳水化合物官能团部分组成，其作为支持的骨架分子。 通常，在质膜表面上发现的糖脂的碳水化合物部分有助于这些分子有助于细胞-细胞识别，通信和细胞间粘附。

complete biocompatibility (Kuo & Prestwich, 2011), having promising biomedical applications (Highley et al., 2016; Larrañeta et al., 2018).

Understanding the factors that influence the properties of HA hydrogels while maintaining their biocompatibility is of great industrial and medical significance (Kuo & Prestwich, 2011). Currently, their design often involves chemical modifications of HA to improve its macroscopic properties (Karel et al., 2018; Yuan et al., 2023) or adding cationic peptides that can promote the cross-linking of HA chains (Barbier et al., 2023; Karel et al., 2018). The latter aspect inspires this work, as peptides can be designed to be biocompatible and can be used to alter the HA behavior substantially, both in solution (Campo et al., 2013; Mummert et al., 2000) or in a hydrogel (Yuan et al., 2023).

Interactions between hyaluronan and proteins are abundant in biological systems (Day & Prestwich, 2002; Garantziotis & Savani, 2019) and were reported as weak (Vuorio et al., 2017) and multivalent (Wolny et al., 2010), requiring several binders per hyaluronan moiety. We could learn the secrets of good HA binders from these natural interactions to later engineer new biocompatible protein mimetics with enhanced or specific interactions. Unfortunately, the complex and dynamic glycocalyx structure entangles experimental and theoretical investigations of hyaluronan in its natural environment. Currently, most known HA binding sites belong to transmembrane receptors, such as the glycoprotein CD44 or TLR4, and their full molecular characterization was achieved only recently with sufficient accuracy in their functional forms (Liu & Finzel, 2014; Vuorio et al., 2021). One of the CD44 residues identified as crucial in CD44–HA interactions is R41, *i.e.*, a positively Arg的重要性 charged arginine. Other arginine residues on the same protein face stabilize some binding modes (Vuorio et al., 2017). This agrees with intuitively accessible electrostatic interactions expected for the negatively charged HA (Giubertoni et al., 2021; Kale et al., 2020; Lenormand et al., 2008, 2010; Sterling et al., 2021). However, there is little evidence of whether electrostatics is the sole driving force in HA–protein recognition and other factors may be at play, like hydrophobic regions (Vuorio et al., 2017). Gaining insights into these patterns is not only fundamental to biology but can also guide the design of natural cross-linking peptides with a desired mode of action and enhanced biocompatibility.

It is also not entirely clear what the typical strength of HA–protein interactions is, *i.e.*, whether HA and protein(s) form stable and relatively static intermolecular complexes or if HA binding has a dynamic nature associated with multiple binding/unbinding events while maintaining interaction selectivity. Molecular dynamics (MD) simulations, which are commonly used to obtain an atomic-resolution view of HA interactions (Nagarajan et al., 2022), do not always present a consistent picture of HA–protein binding (Guvench, 2022). This discrepancy is related to the sensitivity of interaction details to the force field models used in simulations, emphasizing the need to evaluate these models' overall performance carefully. For instance, it has been shown that popular nonpolarizable force fields may struggle to accurately describe carbohydrate–carbohydrate (Lay et al., 2016) and carbohydrate–protein (Lay et al., 2017) interactions, as well as interactions involving charged moieties (Duboué-Dijon et al., 2020). Consequently, previous MD studies that employed these force fields might have obtained a distorted representation of the intermolecular complexes shaped by carbohydrates, especially in combination with amino acids, peptides, and proteins. Therefore, there is a demand for combined experimental and theoretical studies where molecular simulations complement experimental measurements and are validated against them.

This work shows that hyaluronan binding to peptides is sizeable, making them promising biocompatible candidates to alter hyaluronan solutions properties and hydrogels. It is driven by charge complementarity, while we suggest that the molecular specifics of the amino acids' side groups also play a relevant role in the binding strength and structural features, such as the bridging of hyaluronans. We also propose that the interaction between hyaluronan and peptides defies the lock-key paradigm, showing a continuous, dynamic, and large interaction surface covering most of the hyaluronan, especially for arginine, which justifies its enhanced binding strength. We elaborate on these questions by combining NMR measurements with molecular dynamics simulations of hyaluronan–oligopeptide mixtures.

2. Methods

2.1. Reagents

The hyaluronan octamer ($HA8^{AN}$, from now on just HA8) and larger polysaccharide (HA_{long}) with a molecular weight of 15–30 kDa were purchased as a powder from Contipro S. A. The fluorenylmethoxycarbonyl(FMOC)-amino acids were purchased from Iris Biotech. The tetrapeptides (R4, K4, and G4) were synthesized and capped (N-amidated and acetylated at C- and N-termini, respectively) *in house* following a standard solid-phase peptide synthesis procedure (Hansen & Oddo, 2015).

Peptides were synthesized by SPPS on Liberty Blue peptide synthesizer (CEM, USA) using standard Fmoc chemistry protocols, DIC/Oxyma coupling reagents, and Rink Amide MBHA resin support (0.1 mmol scale, 10 equivalents amino acid excess). The crude peptide was dissolved in water and injected on the Vydac protein and peptide C18 reverse phase column 218TP101522. The flow of solvents was 7 ml/min. The elution has been compiled from 20 min washing with water and 0.05 % trifluoroacetic acid, with the gradient increasing 1.25 % methanol per minute. Elution times were 30–36 min, 9–14 min, and 12–19 min, respectively, for R4, K4, and G4. The purity was assessed by analytical RP-HPLC (Vydac 218TP54 column) and LC/MS (Agilent Technologies 6230 ToF LC/MS).

Note that HA8 was chosen for this study as one of the longest HA polymers with a precise length that is readily available at a reasonable price. The length of peptide sequences was consequently chosen to maintain the molar and charge neutrality in the mixtures with HA8.

2.2. NMR measurements

2.2.1. Chemical shift perturbation

The chemical shift perturbation method (Williamson, 2013) was employed to assess possible interactions between hyaluronan and the studied tetrapeptides. The technique is based on the analysis of changes in the chemical shift of one substance upon the titration with another, which indicates possible interactions between them.

We prepared several solutions with varying amounts of HA8 while the concentration of hydrated peptides (R4, K4, G4) was kept constant (1.62 mM). The final peptide/HA8 molar ratios were 1:1, 1.5:1, and 2:1 for R4–H8 and K4–HA8 solutions, and 1:1 and 2:1 for G4–HA8 solutions. We also prepared solutions with HA_{long} (with molar peptide/HA_{long} ratios of 1:2 and 1:4) to verify that the observed trends remain the same irrespective of the molecular weight of the polysaccharides. To rule out the possibility of a concentration effect being the cause of the changes in the spectra, solutions of HA_{long} with the same mass/volume concentrations as in the peptide–HA8 mixtures were measured in the absence of peptides.

All the solutions were prepared in D_2O, and their ^{1}H spectra were collected. The APT ^{13}C, HSQC, HMBC, and COSY spectra were additionally acquired for assignment purposes when necessary (all spectra can be found in the SI). The NMR spectra were recorded on a 400 MHz Bruker AVANCE III spectrometer (^{1}H at 401 MHz, ^{13}C at 101 MHz) equipped with a liquid-nitrogen cryoprobe. The spectra were referenced to the solvent signal $\delta(^1H) = 4.79$ ppm (D_2O). Every spectrum was obtained at 295 K. The resulting spectra were analyzed using the Mnova software (Willcott, 2009). The position of the signals was selected using the Mnova tool "select peaks" to choose the position of the maximum unequivocally.

2.2.2. Water suppression experiments

Water suppression experiments were carried out for pure solutions of hydrated K4, R4, HA8, and their 1:1 mixtures (*i.e.*, K4–HA8, R4–HA8). These experiments are required to identify labile protons like those of amide nitrogens, which are invisible in typical NMR experiments with deuterated water due to hydrogen–deuterium exchange. The peptide and HA8 concentrations were 2.44 mM in all cases, and the solutions were prepared in a 90:10 mixture of H_2O/D_2O. ^1H, ^1H-^{15}N HSQC, HMBC, and NOESY spectra were recorded at ≈ 295 K on a Bruker AVANCE III 600 MHz spectrometer (^1H at 600.1 MHz, ^{13}C at 150.9 MHz, ^{15}N at 60.8 MHz) and/or Bruker AVANCE III 500 MHz spectrometer (^1H at 500.0 MHz, ^{13}C at 125.7 MHz, ^{15}N at 50.7 MHz). The proton spectra measured in the H_2O–D_2O mixture were acquired with water signal suppression using excitation sculpting with gradients (Hwang & Shaka, 1995). The signal assignment is based on homo- and heteronuclear correlation experiments: COSY, HSQC, HMBC, and NOESY.

2.3. Molecular dynamics

2.3.1. Computational models

The hyaluronan octamer (HA8) was built in CHARMM-GUI (Lee et al., 2016; Park et al., 2019), and the tetrapeptides (K4, R4, and G4) were drawn in Avogadro (Hanwell et al., 2012). We also modeled RKRK, A4, P4, and E4 amino acid sequences for additional comparison. The C- and N-termini of each tetrapeptide were capped by N-amide and acetyl groups, respectively, to match the experimentally synthesized moieties. CHARMM-GUI was used to generate model topologies for both HA8 and tetrapeptides.

We employed two MD force fields in this work. The CHARMM36m force field is an all-atom generic biological force field containing parameters for all biomolecules, including proteins and polysaccharides (Guvench et al., 2011; Huang et al., 2016; MacKerell et al., 2010). The prosECCo75 force field (Nencini et al., 2022) is a refined version of CHARMM36m that reintroduces the electronic polarization to the originally nonpolarizable force field *via* charge scaling (Leontyev & Stuchebrukhov, 2011). The only difference between these models is the atomic charges of ions and charged groups, *i.e.*, peptide ammonium and guanidinium groups in side chains and hyaluronan carboxyl groups, see Table S1. The partial atomic charges of these ionic groups were scaled down so that the total charge of each group was 75 % of its original value. The charge scaling reflects the electronic polarization upon ionic solvation and has been shown to improve the quality of MD simulations over a range of biological systems (Duboué-Dijon et al., 2020). Note that the initial choice of CHARMM36m model is dictated by its better ability to describe carbohydrate–amino acid and carbohydrate–carbohydrate interactions (Lay et al., 2016, 2017) as compared to GLYCAM06 (Kirschner et al., 2008), another popular model for carbohydrates.

2.3.2. Simulation protocol

Most of the simulated systems consisted of one hyaluronan octamer and one tetrapeptide, see Fig. S22 for a simulation snapshot of an example system. A 5 nm cubic box with both solutes initially separated by a distance of at least 1 nm was solvated in CHARMM-compatible TIP3P water (Jorgensen et al., 1983). For neutral (*i.e.*, G4, A4, and P4) and negatively charged (E4) peptides, the system was neutralized by four and eight potassium ions, respectively. Additionally, we performed simulations where two R4, K4, or G4 peptides and two HA8 molecules were solvated in a 6.5 nm cubic box, with potassium counterions being added when required. Finally, we also performed additional reference simulations: only hyaluronan octamer with potassium counterions (in a 5 nm cubic box) and only R4, K4, or G4 peptide with chloride counterions when necessary (in a 4 nm cubic box).

All simulations were performed in the *NPT* ensemble. The temperature of 300 K was maintained employing the Nosé–Hoover thermostat with a coupling time of 1 ps (Hoover, 1985; Nosé, 1984). The pressure of 1 bar was controlled *via* the Parrinello–Rahman barostat with a coupling time of 5 ps (Parrinello & Rahman, 1981). The cut-off for electrostatic interactions was set to 1.2 nm, while the particle mesh Ewald method was used to calculate the long-range contribution (Essmann et al., 1995). The Lennard-Jones interactions were smoothly turned off between 1 and 1.2 nm using the force-based switching function (Steinbach & Brooks, 1994). The water geometry was constrained using the SETTLE algorithm (Miyamoto & Kollman, 1992). All other covalent bonds involving hydrogens were constrained using the LINCS algorithm (Hess, 2008; Hess et al., 1997). All simulations were at least 1 μs long; simulations involving R4, K4, or G4, and run with prosECCo75 force field, were prolonged till 2 μs to ensure the convergence and viability of our results. For the same reason, we performed a replica simulation of a representative system (see SI for details). In all cases, the first 100 ns were treated as equilibration, and the rest was used for the analysis. The coordinates were saved every 50 ps. All simulations were performed in GROMACS software, versions 2020.5, 2021.1, and 2022.3 (Abraham et al., 2015). The MD trajectories were analyzed using *in house* Python scripts and post-processing inbox tools in GROMACS. The standard error was estimated by splitting the analysis into five time-equivalent intervals. Simulation input and output files are openly available in Zenodo at DOI: https://doi.org/10.5281/zenodo.8028600.

3. Results and discussion

In this work, we investigate the binding affinity of oligopeptides to hyaluronan using a combination of NMR and MD methods. Hyaluronan has previously been suggested to preferentially interact with positively charged amino acids, as expected from its overall negative charge (Ceely et al., 2023; Chytil et al., 2016; Hong et al., 2022; Jugl & Pekař, 2020; Vuorio et al., 2017). The comparison of collected ^1H NMR spectra, Fig. 1a-c, indicates specific preferences for positively charged peptide sequences. This trend is reflected in notable concentration-dependent changes in chemical shifts upon adding R4 and K4 tetrapeptides to the solution with hyaluronan octamer. At the same time, NMR measurements on G4–HA8 mixtures show negligible alteration of the spectra at every concentration measured.

The MD data perfectly resemble the conclusions from interpreting the NMR results, Fig. 1d. We report the probability of tetrapeptides to interact with hyaluronan from corresponding simulations of tetrapeptide–HA8 solutions. Positively charged tetrapeptides notably prevail in binding affinity over neutrally and negatively charged, as seen from MD simulations with the two employed force field parameterizations. Both models eventually provide practically the same qualitative picture. The only significant difference is an even higher interaction probability for arginine- and lysine-containing tetrapeptides with the CHARMM36m model. This is not surprising because the CHARMM36m model is known to overestimate electrostatic interactions (Duboué-Dijon et al., 2020) when no NBFIX (Non-Bonded Fix) parameters (Yoo & Aksimentiev, 2018) are available for a given charge pair. This is, unfortunately, the case for hyaluronan–peptide binding. Once electrostatic interactions are appropriately screened, as introduced in the prosECCo75 model, the interaction probability for positively charged tetrapeptides decreases but remains notably higher than for other tetrapeptides.

NMR and MD data also point to a considerable difference between arginine and lysine in their affinity for hyaluronan, Figs. 1a-b,d. This discrepancy is nicely seen when comparing the NMR data for peptide–hyaluronan solutions with 1:1 M ratios or the corresponding MD results. Moreover, the MD simulations with the CHARMM36m model suggest that tetraarginine is irreversibly bound to hyaluronan, which is not true for tetralysine. The results for the additionally modeled RKRK tetramer essentially show that the presence of two arginines increases the interaction probability to be almost as high as for tetraarginine, including introducing the irreversible character of interactions in CHARMM36m simulations. This observation also denotes a leading role of arginine in binding hyaluronan. At the same time, NMR data suggest

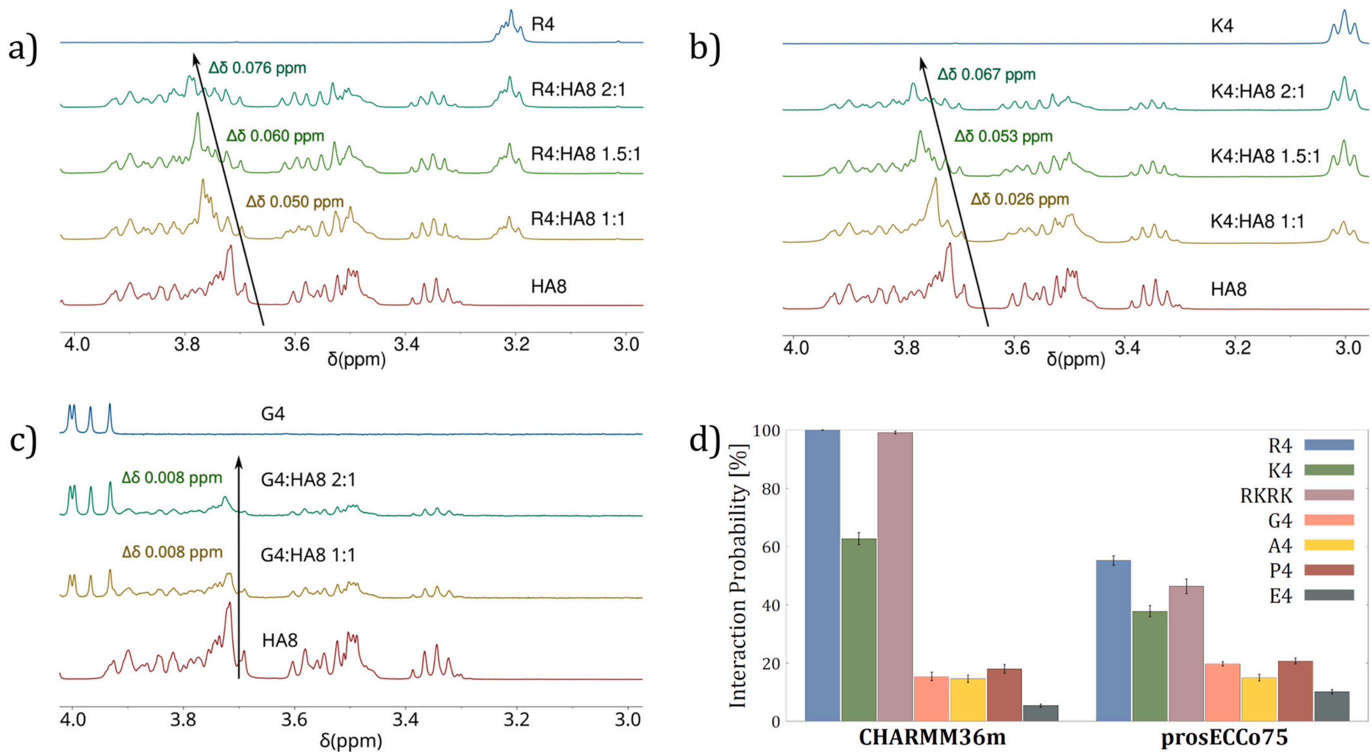

Fig. 1. The [1]H NMR spectra measured on hyaluronan octamer (HA8) solutions with a) R4, b) K4, and c) G4 tetrapeptides. The regions with the largest chemical shift perturbations are shown. The black arrows visually guide through the displacement (or its absence, like in the case of G4) of the chemical shifts upon adding a tetrapeptide to HA8 solution. The corresponding quantified shifts in the position of the signals are also given and colored following the spectra color scheme. d) The interaction probability of HA8 with modeled tetrapeptides calculated from molecular dynamics simulations with two different force field parameterizations. The probability is calculated as the fraction of simulation frames when any atoms of HA8 and tetrapeptide are closer than 0.35 nm.

somewhat similar atomistic interaction patterns for both R4 and K4 as given by qualitatively comparable changes in the NMR spectra of the corresponding peptide–hyaluronan solutions. We do not observe any remarkable difference in relatively weak interactions of A4, G4, and P4 (all neutral tetrapeptides) in either set of MD simulations. These data suggest that the strength of hyaluronan–peptide interaction is dictated mainly by electrostatics (for instance, negatively charged E4 has the lowest binding affinity due to repulsive interactions) and only further governed by the side-chain specificity of basic amino acids.

The [1]H NMR spectra measured on larger HA (HA_{long}) have similar features as those measured on HA8 solutions, see Figs. S1, S2, S3 in the Supporting Information, where all the full spectra are summarized and compared. Upon adding positively charged tetramers, we observe the same signals affected and similar displacement magnitudes consistent with the measurements on tetrapeptide–HA8 mixtures. These results indicate that the binding affinities and interaction patterns are likely independent of polysaccharide molecular weight and peptide–hyaluronan molar ratio. We also confirmed that the chemical shift perturbations are caused by hyaluronan–peptide interactions, not by the change in hyaluronan concentration, see Fig. S4.

Fig. 2a shows the concentration-dependent changes in the chemical shifts for hyaluronan–tetrapeptide mixtures of different molar ratios. The most significant perturbations (the order of 8×10^{-2} ppm) for R4–HA8 and K4–HA8 solutions are observed for the peak at 3.72 ppm. This peak corresponds to either GlcA4, GlcA5, or one of GlcNAc6 protons. The complete signal assignment is provided in Fig. S5. These three protons are indistinguishable by NMR measurements alone, and it is impossible to tell if all or only some of them are affected by the presence of tetrapeptides. Interestingly, all three protons are close to hyaluronan carboxyl groups, see Fig. 2b for a schematic binding pattern. This pattern suggests electrostatic interactions between the carboxyl groups and positively charged side chains of the tetrapeptides, which are stabilized by interactions with neighboring atoms or/and cause perturbations of neighboring protons. The same logic applies to changes in the signal at 2.02 ppm assigned to one of the acetamide (GlcNAc-Ac) protons, also closely located to carboxyl groups in hyaluronan polymers.

The anticipated non-bonded interactions can be extracted from MD simulations as radial distribution functions (RDFs) shown in Fig. 2c. One can see that carboxyl and amide oxygens of hyaluronan (O_{COO^-} and $O_{GlcNAc-Ac}$, respectively) interact with side-chain cation hydrogens of basic amino acids (H_{Guan} and H_{Ammon}), while methyl and methylene hydrogens of hyaluronan (H_{CH_3} of GlcNAc-Ac and H_{CH_2} of GlcNac6) indeed mildly interact with amide oxygens of tetrapeptides ($O_{peptide\ bond}$). All these MD results agree with the interpretation of the NMR data. Moreover, in the case of R4, hyaluronan–tetrapeptide binding can be stabilized by π-interactions of the guanidinium cation (the arginine side chain), see RDFs of guanidinium carbon CZ as compared to the nitrogen NZ atom of ammonium group of lysine and the alpha carbon CA atom of glycine. These interactions are especially prominent for methylene hydrogens, which further explains the largest chemical shift perturbations at 3.72 ppm.

The displacements of signals at 3.34 ppm and 4.45 ppm correspond to chemical shift perturbations of GlcA1 and GlcA2 protons, respectively. Besides direct proximity contact (see below for more details), these changes might also indicate the possible bending of hyaluronan polymer upon interacting with positively charged peptides to optimize their interaction since both protons are close to the GlcA-β(1,3)-GlcNAc bond. Noticeably, the chemical shifts of the previously mentioned GlcA4 and GlcA5 hydrogens also can be altered by changes around the GlcNAc-β(1,4)-GlcA bond. The possible reduction of the hyaluronan rigidity in the presence of R4 and K4 can induce or be induced by the reorientation of acetamide and hydroxymethyl groups. Most force fields, including CHARMM-based models, incorporate quite rigid glycosidic linkage for hyaluronan polymers (Taweechat et al., 2020). Therefore, we do not

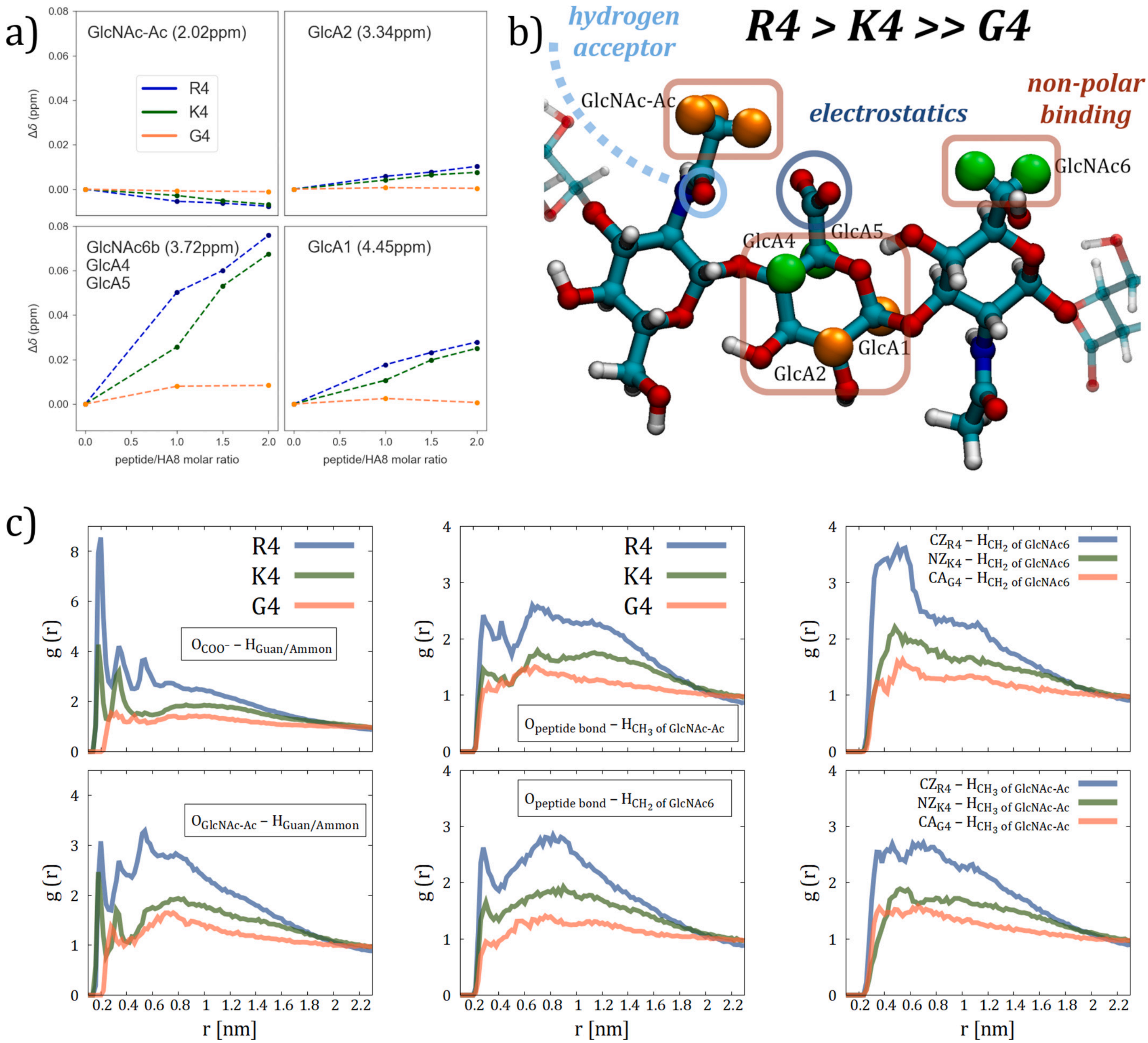

Fig. 2. a) The summary of chemical shift perturbations ($\Delta\delta$) of selected HA8 peaks from ^{1}H NMR spectra measurements after the addition of R4 (blue), K4 (green), or G4 (orange) tetrapeptide to HA8 solution. The data are presented as the difference between the position of the signal maximum in each mixture with respect to the reference, which is the position of the signal in pure HA8 solution (given in brackets). Each signal is assigned to certain HA8 proton(s) as shown in the Supporting Information, Fig. S5. b) The suggested molecular pattern for hyaluronan interactions with studied tetrapeptides. Enlarged colored spheres are hydrogens with the largest chemical shift perturbations measured by NMR. The green enlarged protons correspond to the largest displacement in the NMR signal at 3.72 ppm, and the orange enlarged spheres correspond to the rest of the identified chemical shift perturbations. c) The radial distribution functions for various atom groups of hyaluronan and tetrapeptides from molecular dynamics simulations with the prosECCo75 model.

observe any perceptible bending in our MD simulations, Figs. S23 and S24, except CHARMM36m simulations of HA8–R4 with their utterly strong interactions. Nevertheless, for short HA fragments, the rigidity of glycosidic linkages is rather expected. Additionally, recent studies indicate that the three-dimensional conformation and rigidity of hyaluronan is indeed shaped by glycosidic linkages (Lutsyk & Plazinski, 2021) and not, for instance, by a possible distortion of intramolecular hydrogen bonds, *e.g.*, between acetamide and carboxyl group (Giubertoni, Koenderink, & Bakker, 2019), see Fig. S25, as might be induced by hyaluronan interaction with peptides.

We observe a tendency of charged tetrapeptides to interact with the middle of the hyaluronan chain rather than polymer end-monosaccharides, see Fig. 3b, which reports the relative probability of a tetrapeptide to interact with each monosaccharide of the hyaluronan octamer. In the case of G4, such interaction preference is absent. Interestingly, we generally see a higher interaction probability for amino monosaccharides. This tendency can be explained by the fact that each charge–charge peptide–carbohydrate binding event *via* a glucuronic acid (most often belonging to GlcA3 and GlcA5 monosaccharides) is often stabilized by interactions with two neighboring *N*-acetyl-D-glucosamines, see again Fig. 2b. Another reason is that non-specific intermolecular interactions are simply more probable for larger monosaccharides, *i.e.*, *N*-acetyl-D-glucosamines, as seen in the example of G4.

The peptide side-chain densities surrounding the HA molecule, Fig. 3c, provide further evidence of the superior binding affinity of R4

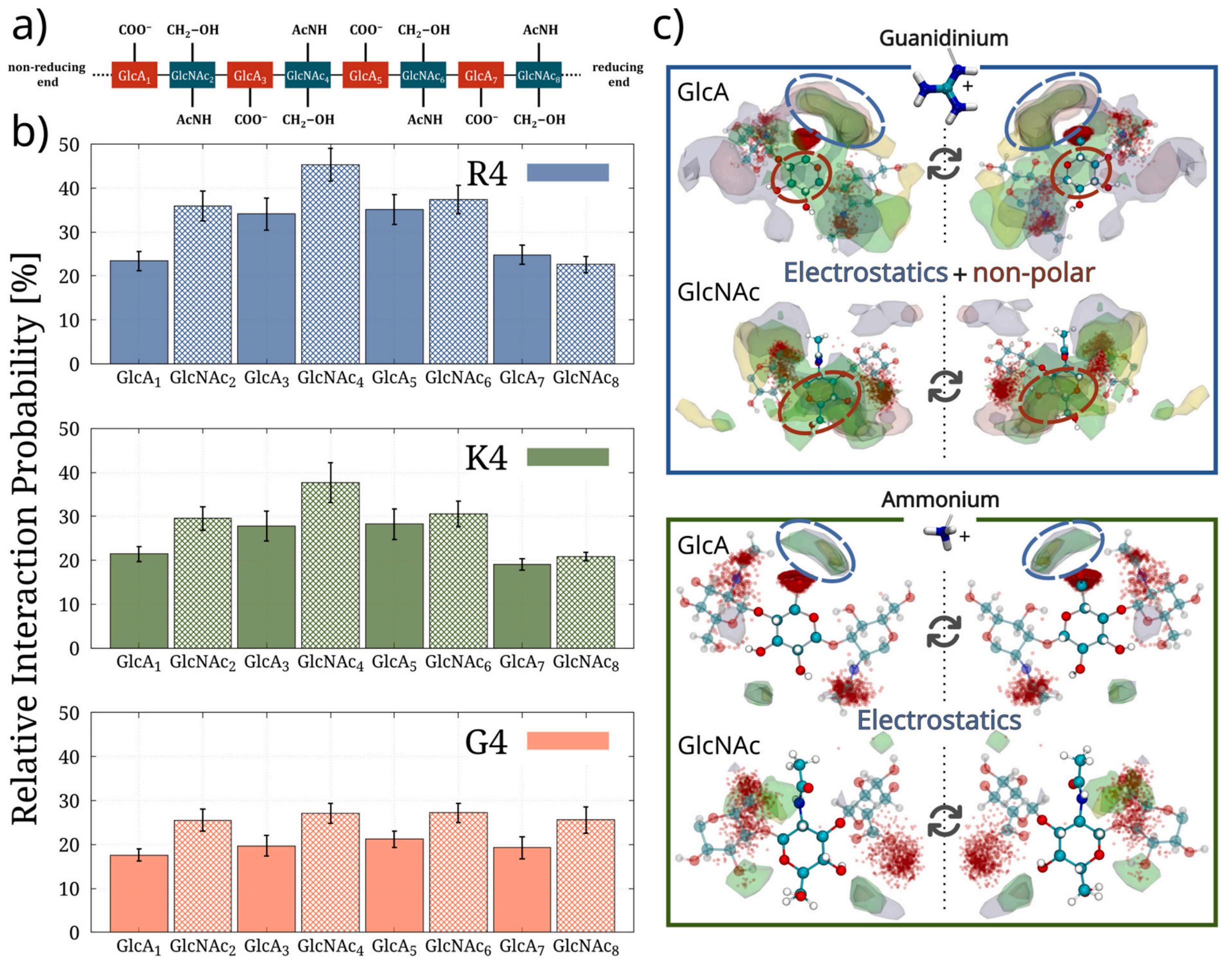

Fig. 3. a) Hyaluronan octamer scheme highlighting its key functional groups. b) Relative percentage of the frames with binding (see Fig. 1d) where the tetrapeptide binds to each hyaluronan monosaccharide as calculated from MD simulations with the prosECCo75 model. The data for the CHARMM36m model is given in Fig. S28. c) Volumetric densities of peptide side chains (guanidinium carbons or ammonium nitrogens) near HA, with the trajectories centered in all ring atoms of either GlcA or GlcNAc monosaccharides. prosECCo75 trajectories were aligned to the six atoms forming the central monosaccharide ring. The isovalue density was set to 0.0017 $Å^{-3}$. Overlayed density maps with different colors correspond to distinct monosaccharides: (yellow) $GlcA_1$ and $GlcNAc_2$, (green) $GlcA_3$ and $GlcNAc_4$, (gray) $GlcA_5$ and $GlcNAc_6$, and (pink) $GlcA_7$ and $GlcNAc_8$. The distribution of coordinates visited by the carboxyl and amide oxygens is represented as red dots. When centering GlcNAc, the distribution of amide oxygen was removed for clarity. Saccharide units flanking the central one correspond to a random configuration from many possibilities.

compared to K4. The presence of densities near the carboxyl and amide oxygen moieties underscores their importance in the interaction, agreeing with our experimental results. This observation reinforces the notion that the carboxyl and acetamide groups are central in bringing the peptide and carbohydrate together and initiating the interaction. The larger dispersion of R4 densities can be attributed to its ability to form non-polar interaction with the more hydrophobic regions of the carbohydrate structure, in contrast to the distribution of K4, predominantly localized around the negatively charged carboxyl group. The establishment of interactions with the carbohydrate surface, in addition to the already mentioned coulombic interactions inherent to charge-unlike molecules, is the primary reason behind the enhanced specificity of arginine-containing peptides.

We performed water suppression NMR experiments to understand the hyaluronan interactions with tetrapeptides further. The water suppression experiments allow us to characterize changes in labile N-protons since in ^{1}H NMR measured in D_2O their signals are invisible due to rapid hydrogen–deuterium exchange. The measurements were carried out for mixtures (*i.e.*, R4–HA8 and K4–HA8 1:1) and pure solutions (*i.e.*, HA8, K4, and R4). Evident modifications in the position and even the shape of HA8 N-bound signals can be observed upon adding either K4 or, R4 on Fig. 4a-b. The displacement of HA8 N-bound signal at 8.06 ppm is almost twice as large after the addition of R4 (4.8×10^{-2} ppm) than after adding K4 (2.5×10^{-2} ppm). This peak corresponds to the nitrogen of the acetamide group, which again points to its principal role in hyaluronan–tetrapeptide binding, as we previously suggested by analyzing the D_2O spectra, *cf.* Fig. 2.

At the same time, we observe almost insignificant shifts for the peptides' signal peaks in the water-suppressed ^{1}H spectra. Somewhat notable perturbations are observed only for the R4 side-chain N-bound protons (signal at 7.20 ppm, on the order of 5×10^{-3} ppm). The signals associated with backbone N-bound protons of both tetrapeptides exhibit negligible changes. Despite supporting previous conclusions about the dominant role of basic side chains, including the higher affinity of arginine, these results indicate that the hyaluronan–peptide interactions do not alter the backbone chain structure of tetrapeptides, or at least this is not detected by our measurements. The Ramachandran plots calculated from MD simulations with prosECCo75 model, Fig. S26, support this experimental observation, *i.e.*, torsion angles are negligibly affected by interactions with hyaluronan. Contrary, CHARMM36m simulations, Fig. S27, point to a notable modification in the torsion distribution of R4 originating from excessively strong interactions.

In contrast to ^{1}H NMR, no signals pointing to peptide–HA8 contacts are recorded by the water-suppressed NOESY spectra (Figs. 4c-d, S9, and

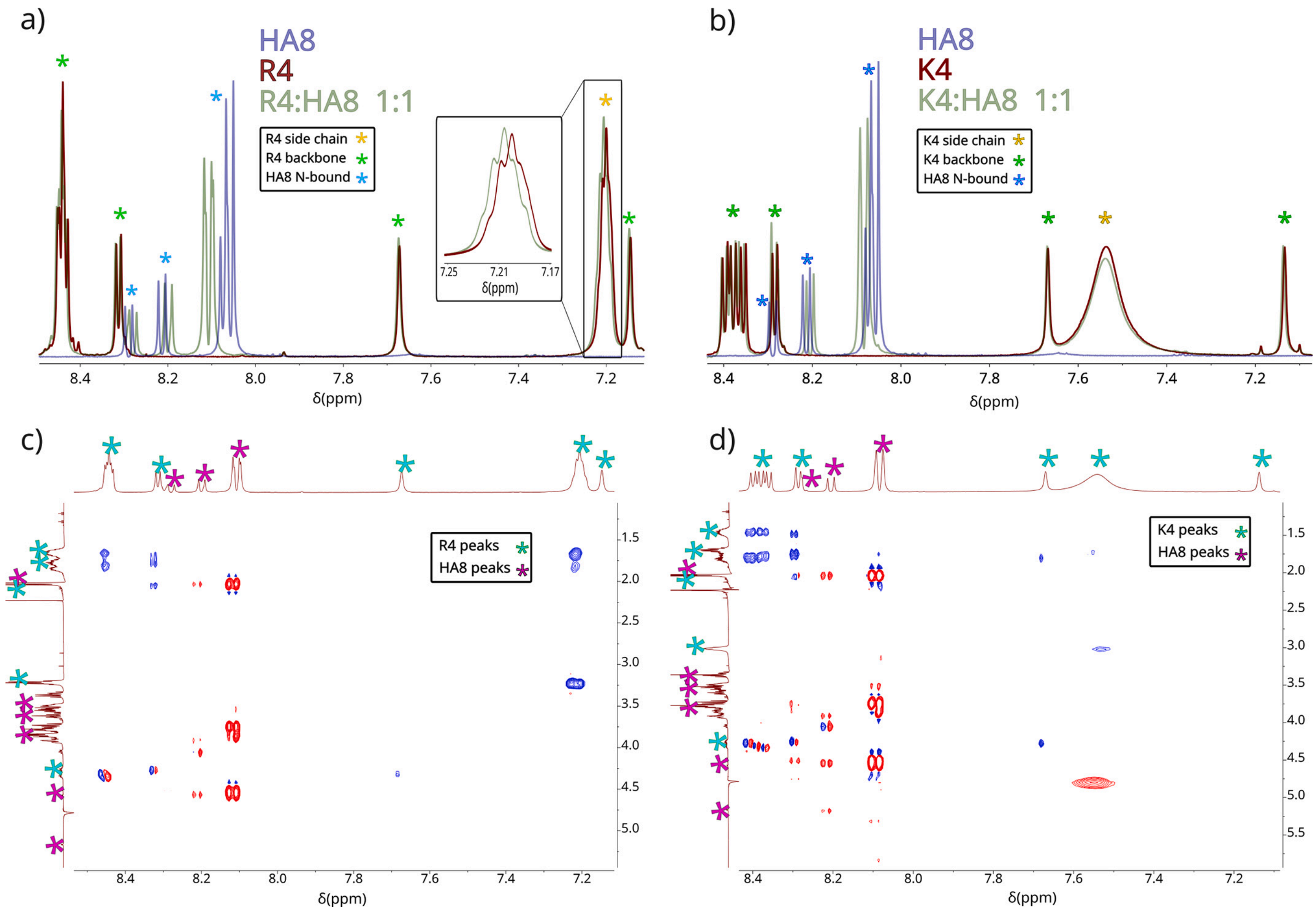

Fig. 4. The top part shows the region of the water-suppressed ^{1}H NMR spectra representing N-bound protons of R4 (a) and K4 (b), respectively, as well as HA8 N-bound protons. The blue lines correspond to the pure HA8 solution, the red lines to pure tetrapeptide solutions, and the green lines represent the tetrapeptide–HA8 1:1 mixtures. The part of the signal with the largest perturbation corresponding to R4 side-chain N-bound protons is magnified and placed into a black box. The NOESY spectra for both R4:HA8 (c) and K4:HA8 (d) solutions from the water suppression experiments are shown in the bottom part. Only the region with NOE signals between N-bound and non-N-bound protons is shown for clarity.

S10). This observation can be explained by a low concentration of bound peptide–HA8 complexes (particularly given the low initial concentration of both solutes in our mixtures) or/and the dynamic mode of tetrapeptide–HA8 interactions, *i.e.*, bound complexes are rather short-lived. Molecular dynamics simulations with the prosECCo75 model support the latter hypothesis. Despite relatively high interaction probabilities for R4 and K4, their binding to hyaluronan is intermittent, without forming shape-defined complexes, and with a lifetime not exceeding dozens of nanoseconds, Fig. 5. Essentially, this dynamic interaction results in multiple possible binding configurations involving 1 to 5 monosaccharides simultaneously, Fig. 5. Contrary, the CHARMM36m model suggests irreversible HA8–R4 binding, Fig. S29, *i.e.*, the formation of an intermolecular complex that does not disassociate on the timescale of our simulations, which does not align with the interpretation of our NMR data. The well-defined and notably stable complexes were also suggested in earlier DFT and MD simulation works (Chytil et al., 2016; Kulik et al., 2023), where conformational ensembles were not properly sampled. The employed GLYCAM06 MD force field is known for producing insoluble carbohydrates–carbohydrates and carbohydrates–peptides complexes (Lay et al., 2016, 2017). This force field, in combination with the short simulation timescales, is inadequate to characterize intermolecular binding. Overall, the previous MD and DFT studies provide the impression of stable and rigid binding because the dynamic nature described here was simply out of their reach (Chytil et al., 2016; Kulik et al., 2023).

Finally, we checked whether the studied peptides together with hyaluronan could form the so-called "bridging" (or "sandwich") complexes, *i.e.*, when one peptide/hyaluronan binds to two hyaluronan/peptide molecules at the same time, respectively. Our additional MD simulations with two peptides (either R4, K4, or G4) and two HA8 molecules, performed using the prosECCo75 model as a more realistic representation of the studied systems, show that both R4 and K4 could be found in such "bridging" configurations, see Table 1, while G4 was not. R4 is more often found in "bridging hyaluronan (3h)" (one hyaluronan binds two peptides) and "bridging peptide (3p)" (one peptide binds two hyaluronan molecules) configurations than K4. More interestingly, the existence of tetramolecular complexes, *i.e.*, all four solutes are interconnected *via* peptide–carbohydrate binding, is substantially more probable in R4–HA8 solutions than in K4–HA8. Consistently with the simulations of the one-to-one systems, the probability of single peptide–single hyaluronan binding is also higher for R4. This suggests that the higher interaction probability, which is clearly observed in our simulations, can translate into a more likely formation of complexes involving multiple molecules. However, once again, the lifetime of such multimolecular complexes was still short, Fig. S30.

4. Conclusions

This work elaborates on the molecular basis of hyaluronan–peptide interactions, combining NMR measurements and molecular dynamics

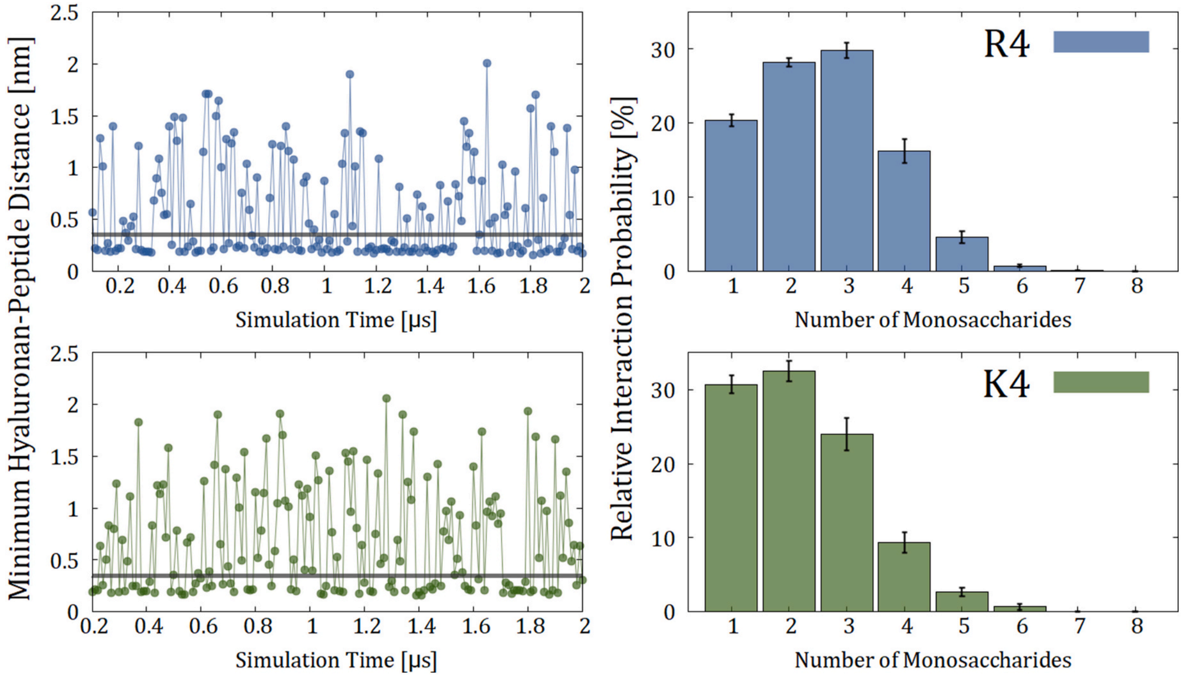

Fig. 5. Time dependence of the minimum distance between hyaluronan octamer and tetrapeptides (left) and the relative probability of tetrapeptide's interaction with a specific number of hyaluronan monosaccharides concurrently (right). The data are shown for HA8–R4 (top) and HA8–K4 (bottom) solutions from molecular dynamics simulations with the prosECCo75 model. The frequency of data points in the minimum-distance graphs is 10 ns, *i.e.*, every 200$^{\mathrm{th}}$ saved simulation frame. The relative probability is calculated as the fraction of simulation frames with at least one contact between tetrapeptide and the corresponding number of monosaccharides with respect to the overall probability of tetrapeptide–hyaluronan binding as given in Fig. 1d.

Table 1
The possibility of the so-called "bridging" binding as seen in MD simulations containing two tetrapeptides (either R4, K4, or G4) and two hyaluronan octamers. The size of the complex is also given, *i.e.*, 4 means that all four solutes are bound to each other, while, *e.g.*, 3h means a trimeric complex, where one hyaluronan (h) binds two peptides. The probabilities are calculated as the fraction of simulation frames with a specific binding type (if any). The data are shown for the prosECCo75 model.

Binding type	Size	R4	K4	G4
Tetramolecular complex	4	18.1±1.9	4.6±0.3	0.6±0.2
Peptide–HA–Peptide trimer	3h	8.1±1.2	3.8±0.4	1.2±0.1
HA–Peptide–HA trimer	3p	4.7±0.9	2.8±0.3	0.3±0.1
Peptide–hyaluronan dimer(s)	2	57.0±2.7	53.0±1.3	27.4±0.8
No peptide–hyaluronan contacts	0	12.1±1.2	35.8±1.2	70.5±1.0

simulations. The arginine-containing peptides were shown to be the superior binders to HA. Despite their identical charge, tetraarginine binds better than tetralysine, thanks to additional non-polar interaction regions that increase the binding strength beyond the necessary electrostatics. These non-polar interactions originate from the partially hydrophobic nature of the guanidinium side chain. This feature, in turn, translates into extended peptide densities surrounding the hydrophobic parts of the carbohydrate. Notably, we do not observe significant changes in the carbohydrate or peptides' internal conformations upon interaction. The binding is dynamic and short-lived, not forming long-standing complexes. The dynamic behavior is experimentally supported by the lack of intermolecular signals in the NOE spectra, and it is only replicated in MD simulations when using force fields that correctly describe electrostatics. Our study points to the importance of selectivity and dynamics in hyaluronan–peptide interactions, opposing the paradigm of "locked" binding modes. Notably, R4 is also the best-suited peptide to bridge different HA chains. This behavior clearly correlates with the fact that R4 is a better binder, and it may also point to its ability to form hydrogels and polymer meshes together with HA. Further experimental and simulation work should address interactions with longer peptide sequences, in combination with an appropriate simulation model, as highlighted in this study.

CRediT authorship contribution statement

Miguel Riopedre-Fernandez: Conceptualization, Methodology, Investigation, Formal analysis, Visualization, Writing – Review and Editing.
Denys Biriukov: Conceptualization, Methodology, Investigation, Formal Analysis, Visualization, Writing - Original Draft.
Martin Dračínský: Investigation NMR, Writing - Review and Minor Editing.
Hector Martinez-Seara: Conceptualization, Methodology, Resources, Supervision, Project administration, Funding acquisition, Writing - Review and Editing.

Declaration of competing interest

The authors declare that they have no financial interests/personal relationships which may be considered as potential competing interests.

Data availability

All simulation data have been uploaded to the public scientific repository Zenodo (DOI: https://doi.org/10.5281/zenodo.8028600).

Acknowledgements

M.R.-F. also acknowledges the support from the Charles University in Prague and the International Max Planck Research School in Dresden. We acknowledge Grammarly and ChatGPT for improving readability and language.

Funding

M.R.-F. and H.M.-S. acknowledge the support of the Czech Science Foundation (project 19-19561S).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.carbpol.2023.121568.

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